

Team Name : Stars

Project Name : SkyInsight

Project Description :

SkyInsight is a full-stack, production-grade AI platform built for the airline industry. It ingests raw operational and customer data, processes it through a structured SQL Server data warehouse, runs it through multiple machine learning pipelines, and surfaces insights via a modern Next.js web application — all in a single, coherent system.

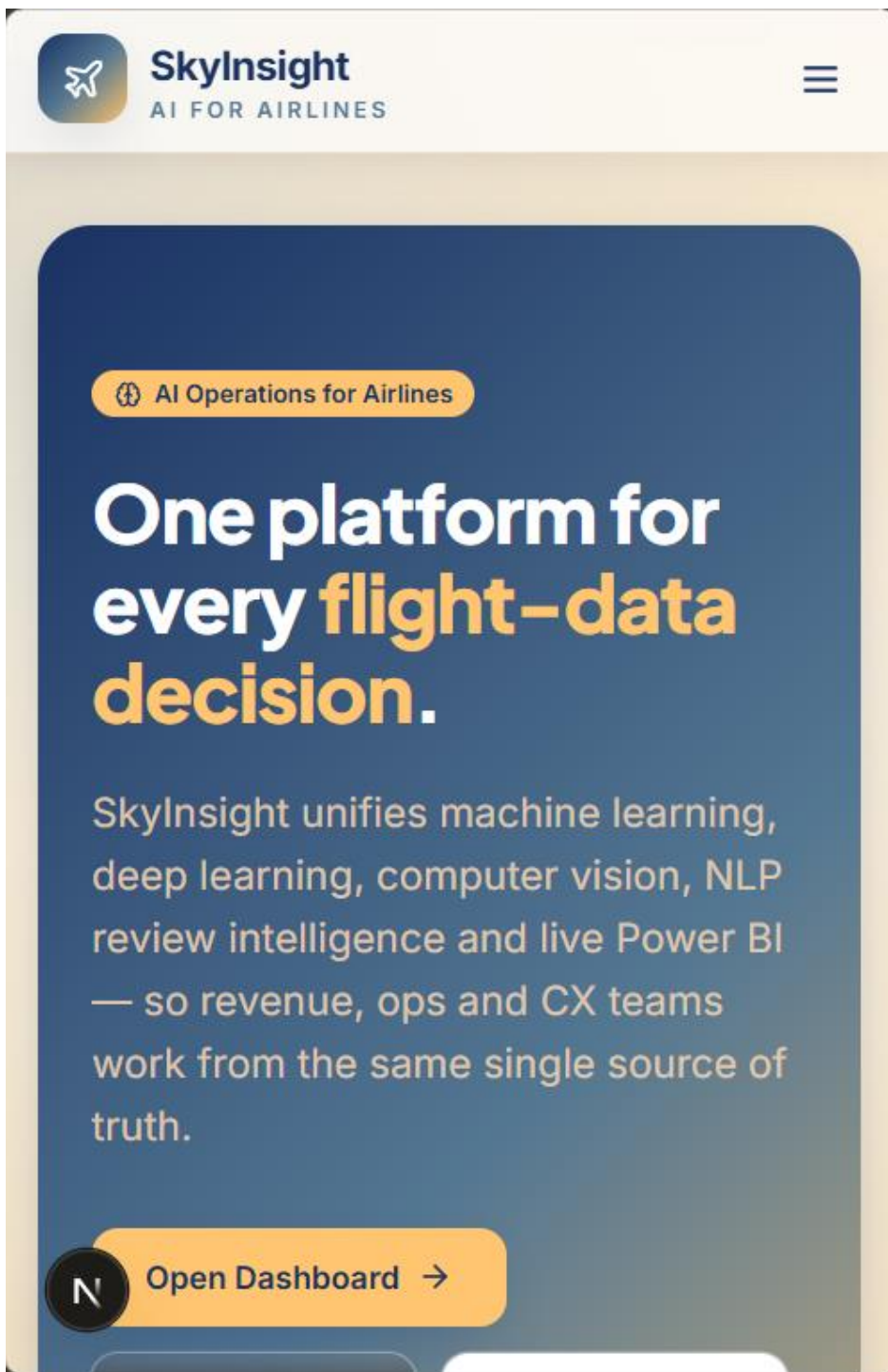
The platform bridges the gap between raw data and real decisions. Whether an analyst needs to understand why passengers are dissatisfied, a ground manager needs to predict the next delay, or a sustainability lead needs to track carbon output per route — SkyInsight delivers the answer in one place, in real time.

The system covers the full AI lifecycle: data ingestion and cleaning → feature engineering → model training → ONNX export → FastAPI serving → interactive front-end. Every layer is built and owned within the project, with no black-box third-party ML services.

Logo (1080 x 1080)



Cover Image for the mobile App (1080 x 720 or 1920 x 1080)



Flyer to be used for marketing materials (1748 x 2480 or 2480 x 3508)



SkyInsight

AI-POWERED AIRLINE INTELLIGENCE PLATFORM

A full-stack analytics and machine learning platform that transforms raw airline data into **actionable intelligence** — spanning customer experience, operational safety, sustainability, and strategic decision-making.

8

AI MODULES

5

ML MODELS

3

CNN CLASSIFIERS

129K+

RECORDS ANALYSED

Real-Time

ONNX INFERENCE

3-Tier

ARCHITECTURE

CORE MODULES



Decision Insights

Interactive BI dashboards surfacing KPIs, route performance, and loyalty segmentation directly from the airline data warehouse.



Satisfaction Prediction

Logistic regression ONNX model predicts passenger satisfaction probability from 22 service and flight features in real time.



Delay Prediction

ANN-based classifier forecasts flight delay risk using route, aircraft type, and scheduling signals with confidence scoring.



Loyal Circle

K-Means customer clustering by loyalty tier and churn risk. Radar charts and profile cards reveal four distinct traveller archetypes.



CO₂ Forecasting

Time-series regression model predicts carbon emissions per route, enabling airlines to optimise for sustainability targets.



Smart Vision

EfficientNetV2-S CNN classifiers detect luggage damage, cabin cleanliness issues, and crowd density from uploaded photos.



Reviews NLP

Web scraping → VADER & RoBERTa sentiment analysis



Performance Hub

Consolidated model metrics, business impact scores

Images of the application or solution

AI Operations for Airlines

One platform for every flight-data decision.

SkyInsight unifies machine learning, deep learning, computer vision, NLP review intelligence and live Power BI — so revenue, ops and CX teams work from the same single source of truth.

Open Dashboard →

View Power BI

Try a live model

8+

ML & DL MODELS

94%

AVG. TEST ACCURACY

30k+

REVIEWS ANALYSED

<50ms

END-TO-END LATENCY

throughs and bookmarks all work.

Live report

Refresh

Fullscreen

Open in Power BI

Executive Dashboard

إشعار بالطلبات

Loyalty Program

Customer Retention & Rewards

Flight Performance

Delays · On-Time · Operations

Passenger Satisfaction

Experience & Feedback

Environmental Reporting

Emissions · Sustainability · ESG

Efficiency & Profit

Revenue · Costs · Margins

Operations · Customer · Loyalty · Environmental · Revenue & Profit

Landing Page

Loyalty Program

Flight Performance

Passenger Satisfaction

Environmental Reporting

Airline Route Profitability Analysis

User Guide Loyalty

User Guide P

COMPUTER VISION

Cabin · Crowd · Luggage

EfficientNetV2S backbone with TTA + temperature-scaled probabilities.

Cabin Cleanliness

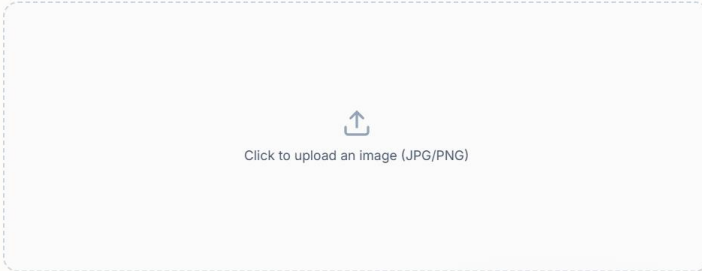
Detect Clean vs Dirty cabins.

Terminal Crowd

Estimate crowd density level.

Luggage Damage

Spot damaged vs good luggage.



Click to upload an image (JPG/PNG)

Classify image

Result

Upload an image and run.

REVIEW INTELLIGENCE

Airline Reviews — NLP Workspace

Scraped airline reviews enriched with VADER, DistilBERT sentiment, LDA topics and spaCy entities.

REVIEWS ANALYSED

3,328



POSITIVE SHARE

72.0%



NEGATIVE SHARE

26.4%



TOPICS DISCOVERED

8



Live sentiment analyser

Run any review text through VADER (lexicon) and DistilBERT (transformer) in one click.

Paste a review... e.g. 'The crew was friendly but my baggage was delayed by 2 hours.'

0 chars

Analyse

Sentiment mix (VADER)



● positive ● negative ● neutral



Commercial video (Max 3 minutes)

List of themes associated with the Project

For **SDG 9 (Industry, Innovation & Infrastructure)**, the system improves airline efficiency and infrastructure resilience through smart route optimisation, airport crowd analytics, delay prediction, and real-time operational dashboards.

For **SDG 12 (Responsible Consumption & Production)**, the platform promotes efficient resource usage by monitoring fuel consumption, optimising cabin cleaning operations, improving baggage handling quality control, and enhancing transparency through NLP-based customer feedback analysis.

For **SDG 13 (Climate Action)**, the solution helps reduce environmental impact by predicting carbon emissions, tracking emission trends, identifying high-emission routes, and providing sustainability reporting through Power BI dashboards. Overall, the platform contributes to smarter, more sustainable, and environmentally responsible airline operations.

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